

10 (short)

(a) Explain the physical basis of Young's modulus for:

- (i) glass,
- (ii) polyethylene.

For which of these materials can Young's modulus be manipulated significantly via processing? Explain your reasoning. [4]

(b) Foaming is a popular method for manipulating elastic properties of solids. A simple model predicts Young's modulus of a foam, E_f , scales with that of the solid E_s as:

$$E_f = E_s \left(\frac{\rho_f}{\rho_s} \right)^2$$

where ρ_f and ρ_s are the density of the foam and the solid respectively.

- (i) In the model, which mechanism is assumed to dominate the elastic response of this foam? [2]
- (ii) A foam made of Nickel has Young's modulus 1.3 GPa. Bulk Nickel has density $\rho_s = 8.9 \text{ Mg m}^{-3}$ and Young's modulus $E_s = 200 \text{ GPa}$. Find the density of this foam. [2]
- (iii) Using the appropriate material property chart in the data book, identify an existing material with which the Nickel foam from part (ii) appears to compete, in terms of modulus and density. Comment on one other property that could determine the suitability of replacing this material with the Nickel foam. [2]